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DBMS MINI PROJECT REPORT AND CASE STUDY

PET CLINIC MANAGEMENT SYSTEM

A Relational Database Application for Veterinary Clinic Records

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1. Introduction

The Pet Clinic Management System is a relational database application developed to organize the routine recordkeeping activities of a veterinary clinic. A clinic handles information about pet owners, animals, veterinarians, appointments, treatments and payments. When these details are maintained in paper registers or separate files, records can be duplicated, misplaced or difficult to retrieve.

This project uses MySQL to store clinic information in related tables. Each table represents a specific entity, and primary and foreign keys connect the records while helping preserve data consistency. SQL queries retrieve information across tables, summarize records and update payment details. The database provides a structured foundation for maintaining clinic records and reviewing a pet's appointment and treatment history.

The implementation is a database-focused prototype. It demonstrates the core data-management functions of a pet clinic and can be extended with a graphical interface, user authentication, reminders and medicine inventory features.

2. Problem Statement and Objectives

2.1 Problem Statement

A veterinary clinic needs reliable access to accurate information about owners, pets, veterinary staff, visits, treatment details and bills. Manual recordkeeping makes it difficult to connect these records, review previous treatments, identify scheduled appointments and monitor unpaid charges. The problem addressed by this project is the need for a centralized database that organizes these related records and allows them to be retrieved and maintained through SQL.

2.2 Objectives

- Maintain contact information for pet owners.
- Store identifying and descriptive details for each pet.
- Maintain veterinarian details and areas of specialization.
- Record appointment dates, reasons for visits and appointment status.
- Store diagnoses, treatment descriptions, medicines and treatment dates.
- Record consultation charges and payment status.
- Use primary keys and foreign keys to link related records.
- Retrieve useful clinic information using SELECT and JOIN queries.
- Demonstrate record updates and aggregate queries in a relational database.

3. System Modules

Module	Function
Pet Owner Management	Maintains owner names, phone numbers, email addresses and addresses.
Pet Management	Stores pet names, species, breeds, dates of birth, gender and owner references.
Veterinarian Management	Maintains veterinarian names, specializations and contact numbers.
Appointment Management	Links pets with veterinarians and stores appointment date, reason and status.
Treatment Management	Records diagnosis, treatment description, medicine and treatment date.
Billing Management	Stores the amount charged, payment status and payment date for an

	appointment.

4. Tools and Technologies

Tool / Technology	Role in the Project
MySQL	Relational database management system used to create and maintain the database.
SQL	Used to define tables, insert data, retrieve information and update records.
MySQL Workbench	Graphical environment for running SQL statements and viewing query results.
Microsoft Word	Used to prepare and format this project report.

4.1 DBMS Concepts Used

The project applies relational table design, data types, NOT NULL constraints, AUTO_INCREMENT identifiers, primary keys, foreign keys, joins, filtering, ordering, grouping, aggregate functions and data updates. A primary key uniquely identifies a row. A foreign key links a row to a valid record in another table. JOIN queries combine related information without storing repeated owner, pet or veterinarian details in every table.

5. Database Design

Database name: pet_clinic_db

Table	Primary Key	Important Relationship
Pet_Owners	owner_id	One owner may be associated with several pets.
Pets	pet_id	owner_id references Pet_Owners(owner_id).
Veterinarians	vet_id	One veterinarian may attend several appointments.
Appointments	appointment_id	pet_id references Pets; vet_id references Veterinarians.
Treatments	treatment_id	appointment_id references Appointments.
Bills	bill_id	appointment_id references Appointments and is unique in Bills.

The database uses a one-to-many relationship between pet owners and pets, pets and appointments, veterinarians and appointments, and appointments and treatment records. In this implementation, each appointment can have at most one bill because appointment_id is declared UNIQUE in the Bills table.

6. SQL Implementation

The following SQL creates the six related tables and inserts sample clinic records. The sample data is illustrative and is included to demonstrate the database operations in this report.

```
CREATE DATABASE IF NOT EXISTS pet_clinic_db;
USE pet_clinic_db;

CREATE TABLE Pet_Owners (
    owner_id INT AUTO_INCREMENT PRIMARY KEY,
    owner_name VARCHAR(100) NOT NULL,
    phone VARCHAR(20) NOT NULL,
    email VARCHAR(100),
```

```
    address VARCHAR(200)
);

CREATE TABLE Pets (
    pet_id INT AUTO_INCREMENT PRIMARY KEY,
    pet_name VARCHAR(60) NOT NULL,
    species VARCHAR(40) NOT NULL,
    breed VARCHAR(60),
    date_of_birth DATE,
    gender VARCHAR(10),
    owner_id INT NOT NULL,
    FOREIGN KEY (owner_id) REFERENCES Pet_Owners(owner_id)
);

CREATE TABLE Veterinarians (
    vet_id INT AUTO_INCREMENT PRIMARY KEY,
    vet_name VARCHAR(100) NOT NULL,
    specialization VARCHAR(100),
    phone VARCHAR(20)
);

CREATE TABLE Appointments (
    appointment_id INT AUTO_INCREMENT PRIMARY KEY,
    pet_id INT NOT NULL,
    vet_id INT NOT NULL,
    appointment_date DATETIME NOT NULL,
    reason VARCHAR(200),
    status VARCHAR(20) DEFAULT 'Scheduled',
    FOREIGN KEY (pet_id) REFERENCES Pets(pet_id),
    FOREIGN KEY (vet_id) REFERENCES Veterinarians(vet_id)
);

CREATE TABLE Treatments (
    treatment_id INT AUTO_INCREMENT PRIMARY KEY,
    appointment_id INT NOT NULL,
    diagnosis VARCHAR(200) NOT NULL,
    treatment_description VARCHAR(250),
    medicine VARCHAR(150),
    treatment_date DATE NOT NULL,
    FOREIGN KEY (appointment_id) REFERENCES Appointments(appointment_id)
);

CREATE TABLE Bills (
    bill_id INT AUTO_INCREMENT PRIMARY KEY,
    appointment_id INT NOT NULL UNIQUE,
    amount DECIMAL(10,2) NOT NULL,
    payment_status VARCHAR(20) DEFAULT 'Unpaid',
    payment_date DATE,
    FOREIGN KEY (appointment_id) REFERENCES Appointments(appointment_id)
);

INSERT INTO Pet_Owners (owner_name, phone, email, address) VALUES
('Aman Sharma', '9876501234', 'aman@example.com', 'Mohali'),
('Simran Kaur', '9876505678', 'simran@example.com', 'Chandigarh'),
('Rahul Verma', '9876509012', 'rahul@example.com', 'Kharar');

INSERT INTO Pets (pet_name, species, breed, date_of_birth, gender, owner_id) VALUES
('Bruno', 'Dog', 'Labrador', '2021-04-12', 'Male', 1),
('Milo', 'Cat', 'Persian', '2022-08-05', 'Male', 2),
('Coco', 'Dog', 'Pug', '2020-11-20', 'Female', 3);

INSERT INTO Veterinarians (vet_name, specialization, phone) VALUES
('Dr. Neha Singh', 'General Veterinary Care', '9811100001'),
('Dr. Arjun Mehta', 'Dermatology', '9811100002');

INSERT INTO Appointments
(pet_id, vet_id, appointment_date, reason, status) VALUES
(1, 1, '2026-10-15 10:00:00', 'Routine health checkup', 'Completed'),
(2, 2, '2026-10-16 11:30:00', 'Skin irritation', 'Completed'),
(3, 1, '2026-10-30 09:30:00', 'Vaccination', 'Scheduled');

INSERT INTO Treatments
(appointment_id, diagnosis, treatment_description, medicine, treatment_date) VALUES
```

```
(1, 'Healthy', 'Routine physical examination',
 'Multivitamin supplement', '2026-10-15'),
(2, 'Mild skin irritation', 'Skin examination and care advice',
 'Vet-approved skin ointment', '2026-10-16');

INSERT INTO Bills (appointment_id, amount, payment_status, payment_date) VALUES
(1, 800.00, 'Paid', '2026-10-15'),
(2, 1200.00, 'Unpaid', NULL),
(3, 600.00, 'Unpaid', NULL);
```

7. Query Results

The query results below show the expected records produced by the sample data above. They are presented as formatted result tables to make the database output clear and readable.

7.1 Pet Owners

```
SELECT * FROM Pet_Owners;
```

owner_id	owner_name	phone	email	address
1	Aman Sharma	9876501234	aman@example.com	Mohali
2	Simran Kaur	9876505678	simran@example.com	Chandigarh
3	Rahul Verma	9876509012	rahul@example.com	Kharar

7.2 Pets and Their Owners

```
SELECT p.pet_id, p.pet_name, p.species, p.breed, o.owner_name, o.phone
FROM Pets p
JOIN Pet_Owners o ON p.owner_id = o.owner_id;
```

pet_id	pet_name	species	breed	owner_name	phone
1	Bruno	Dog	Labrador	Aman Sharma	9876501234
2	Milo	Cat	Persian	Simran Kaur	9876505678
3	Coco	Dog	Pug	Rahul Verma	9876509012

7.3 Appointment Schedule

```
SELECT a.appointment_id, p.pet_name, v.vet_name,
       a.appointment_date, a.reason, a.status
FROM Appointments a
JOIN Pets p ON a.pet_id = p.pet_id
JOIN Veterinarians v ON a.vet_id = v.vet_id
ORDER BY a.appointment_date;
```

ID	Pet	Veterinarian	Date and Time	Reason	Status
1	Bruno	Dr. Neha Singh	2026-10-15 10:00	Routine health checkup	Completed
2	Milo	Dr. Arjun Mehta	2026-10-16 11:30	Skin irritation	Completed
3	Coco	Dr. Neha Singh	2026-10-30 09:30	Vaccination	Scheduled

7.4 Treatment History

```
SELECT p.pet_name, t.diagnosis, t.treatment_description,
       t.medicine, t.treatment_date
FROM Treatments t
JOIN Appointments a ON t.appointment_id = a.appointment_id
JOIN Pets p ON a.pet_id = p.pet_id
ORDER BY t.treatment_date;
```

Pet	Diagnosis	Treatment	Medicine	Date
Bruno	Healthy	Routine physical examination	Multivitamin supplement	2026-10-15
Milo	Mild skin irritation	Skin examination and care advice	Vet-approved skin ointment	2026-10-16

7.5 Bills and Payment Status

```
SELECT b.bill_id, p.pet_name, b.amount, b.payment_status, b.payment_date
FROM Bills b
JOIN Appointments a ON b.appointment_id = a.appointment_id
JOIN Pets p ON a.pet_id = p.pet_id
ORDER BY b.bill_id;
```

bill_id	pet_name	amount	payment_status	payment_date
1	Bruno	800.00	Paid	2026-10-15
2	Milo	1200.00	Unpaid	NULL
3	Coco	600.00	Unpaid	NULL

7.6 Number of Pets by Species

```
SELECT species, COUNT(*) AS total_pets
FROM Pets
GROUP BY species;
```

species	total_pets
Dog	2
Cat	1

7.7 Updating a Payment Record

```
UPDATE Bills
SET payment_status = 'Paid', payment_date = '2026-10-16'
WHERE bill_id = 2;

SELECT bill_id, amount, payment_status, payment_date
FROM Bills
WHERE bill_id = 2;
```

bill_id	amount	payment_status	payment_date
2	1200.00	Paid	2026-10-16

The UPDATE statement changes bill 2 from Unpaid to Paid and records the payment date. The SELECT statement then displays the updated record.

8. Conclusion

The Pet Clinic Management System demonstrates how a relational database can organize essential veterinary clinic records. The database stores pet owners, pets, veterinarians, appointments, treatments and bills in separate tables connected through primary and foreign keys. This design reduces unnecessary repetition and allows related information to be retrieved through SQL joins.

The implemented queries display owner and pet information, appointment schedules, treatment history and billing status. Aggregate queries summarize the number of pets by species, while an UPDATE statement demonstrates how a payment record can be maintained. The project meets its database-focused objectives and provides a foundation for a more complete clinic application.

Future enhancements could include a graphical user interface, staff login and access control, vaccination reminders, medicine inventory management, appointment notifications and printable invoices.